CIS 5270 Midterm 2 Cheatsheet

Lec 14–20: Explainability, LIME/SHAP, Counterfactuals, Data Attribution, Neurosymbolic, RL, RLHF

L14: Explainability Basics

Why? Debug, trust, recourse, compliance (GDPR "right to explanation"), fairness audits, detect spurious correlations (husky-vs-wolf using snow background).

Taxonomy:

- **Intrinsic vs. post-hoc**: inherently interpretable (linear, tree, shallow rules) vs. explained after training.
- **Global vs. local**: whole model vs. single prediction.
- **Model-specific vs. agnostic**: needs internals vs. only query access.

Local fidelity: explainer g approximates black-box f *accurately in a neighborhood* of x: $g(z) \approx f(z)$ for z near x. Desired because a faithful *local* story actually matches f near x; a globally-simple-but-locally-wrong explanation misleads users about this specific decision.

Desiderata: fidelity, interpretability (human-parseable), stability (similar $x \Rightarrow$ similar explanation), completeness, class-discriminativeness.

Saliency / Input Gradient: for class c, map is $\partial f_c(x)/\partial x$. Pixel i = sensitivity of logit f_c to pixel i. *Not* weight updates, *not* avg training pixels, *not* LOO retraining. (MCQ #2 = B.) Issues: noisy, not class-discriminative, fails Adebayo sanity checks (looks similar with randomized weights).

Improvements:

- **SmoothGrad**: avg gradients over Gaussian-noised x. Reduces visual noise.
- **Integrated Gradients (IG)**: $\int_0^1 \nabla f_c(x' + \alpha(x - x'))d\alpha \cdot (x - x')$, path integral from baseline x' to x. Satisfies *completeness*: attributions sum to $f(x) - f(x')$.
- **Grad-CAM**: gradient-weighted last-conv activations. Class-discriminative, low-res.
- **Guided Backprop**: only positive gradients + positive activations.

L15: LIME

Idea: fit simple interpretable g locally around x.

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

$\pi_x(z)$: proximity kernel $\exp(-D(x,z)^2/\sigma^2)$. Ω: complexity penalty. G: linear models / small trees.

Recipe:

1. Perturb $x \to z_1, \ldots, z_n$ (mask superpixels / drop words / sample features). Interpretable binary space $z' \in \{0,1\}^{d'}$.
2. Get labels $y_i = f(z_i)$, weights $w_i = \pi_x(z_i)$.
3. Fit weighted lin model: $\hat{\beta} = \arg\min_\beta \sum w_i (y_i - z_i'^\top \beta)^2 + \lambda \|\beta\|_1$.
4. Coefficients $\hat{\beta}$ = feature importances.

Variants:

- **Linear LIME**: no sparsity ($\Omega = 0$). Dense coeffs.
- **LASSO LIME**: ℓ_1 penalty $\to$ sparse.
- **K-LASSO**: LASSO path picks top-K features, refit unpenalized OLS on those K. Exactly K features, unbiased magnitudes.

Pros: model-agnostic, intuitive, any data type. **Cons:** unstable (reruns give different results), kernel bandwidth σ sensitive, local $\neq$ global, perturbations may go off-manifold.

L15: SHAP / Shapley Values

Shapley value for feature i:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} \big[v(S \cup \{i\}) - v(S)\big]$$

Avg marginal contribution over orderings. $v(S) = \mathbb{E}[f(x) \mid x_S]$.

Four Axioms (memorize!):

- **Efficiency**: $\sum_i \phi_i = f(x) - \mathbb{E}[f(X)]$. *(MCQ #1 = C.)* Attributions explain the gap from average prediction.
- **Symmetry**: if $v(S \cup \{i\}) = v(S \cup \{j\}) \forall S$, then $\phi_i = \phi_j$.
- **Null Player (Dummy)**: if $v(S \cup \{i\}) = v(S) \forall S$, then $\phi_i = 0$.
- **Linearity**: $\phi_i(\alpha v_1 + v_2) = \alpha \phi_i(v_1) + \phi_i(v_2)$.

Shapley is the *unique* attribution satisfying all four.

Conditional vs. Marginal:

$$\text{true: } \mathbb{E}[f(X) \mid X_S = x_S] = \int f(x_S, x_{\bar{S}}) p(x_{\bar{S}} \mid x_S) dx_{\bar{S}}$$

$$\text{SHAP uses: } \int f(x_S, x_{\bar{S}}) p(x_{\bar{S}}) dx_{\bar{S}} \text{ (indep.)}$$

- **Benefit (SA #3a):** no need to model $p(x_{\bar{S}} \mid x_S)$; just sample $x_{\bar{S}}$ from empirical marginal. Tractable, fast, no density estimation.
- **Drawback (SA #3b):** if features correlated, evaluates f at off-manifold points (pregnant+male, height=7ft weight=100lb) $\to$ unreliable f, misleading attributions. Can give nonzero ϕ to features f doesn't actually use.

KernelSHAP: weighted lin regression w/ Shapley kernel $\to$ LIME-shape that converges to Shapley. **TreeSHAP:** exact poly-time for tree ensembles. **DeepSHAP:** SHAP + DeepLIFT for nets.

L16: Counterfactual Explanations

Counterfactual x': minimally changed input s.t. $f(x') \neq f(x)$. Answers: "what would I need to change?" $\to$ **recourse**.

Minimum-Distance CF:

$$x' = \arg\min_{x'} d(x, x') \text{ s.t. } f(x') = y'$$

Wachter relaxation: $\min_{x'} d(x, x') + \lambda \ell(f(x'), y')$.

Distances:

- L_1 Manhattan $\sum_i |x_i - x_i'|$: *sparse* (few features).
- L_2 Euclidean: smooth, many features.
- L_0: count of changed features (NP-hard).

Critical assumption for L_1/L_2 (MCQ #4 = C): features on **same scale**. If income is in \$ and age in years, L_1 dominated by income. Standardize / MAD-weighted distance. *Not* independence, *not* uncorrelated, *not* equal variance—**scale** is the issue.

Why CFs fail (SA #4):

- **Actionability**: "change age 30→45"—can't.
- **Immutability**: race, gender, birthplace fixed.
- **Unrealistic**: "income +\$500k while cashier".
- **Causal violations**: changes one feature, ignores downstream (income without job change).
- **Not diverse**: one CF may be infeasible.

Example answer: "Raise credit score 580→780 for loan approval." Minimal in L_1 but impossible short-term—useless recourse.

Desirable CF: validity, proximity, sparsity, actionability, plausibility (on-manifold), diversity (DiCE), causal consistency.

Concept-Based (TCAV):

- Define concept by examples ("stripes"); random counter-set.
- Train linear probe on layer-l activations: concept vs random.
- Normal to boundary = **CAV** v_C^l.
- **Directional deriv** $S_{C,k,l}(x) = \nabla h_{l,k}(f_l(x)) \cdot v_C^l$.
- **TCAV score**: fraction of class-k inputs where $S > 0$. Significance-tested vs random concepts.
- Interprets in human concepts ("stripes," "dotted"), not pixels.

L17: Data Attribution

Goal: which train points caused this prediction? Uses: debugging, data valuation, mislabel detection, unlearning, memorization.

Leave-One-Out (LOO): retrain without z_i, observe Δ prediction. Ground truth but **impractical** (MCQ #3 = B): needs n retrainings. *Not* impossible, *not* accuracy claim, *not* re-training deployed nets.

Influence Functions (Koh & Liang, ICML '17):

$$\mathcal{I}(z_i, z_{\text{test}}) = -\nabla_\theta L(z_{\text{test}}, \hat{\theta})^\top H_{\hat{\theta}}^{-1} \nabla_\theta L(z_i, \hat{\theta})$$

$H_{\hat{\theta}} = \frac{1}{n} \sum_j \nabla_\theta^2 L(z_j, \hat{\theta})$. Upweighting z_i by ϵ changes $\hat{\theta}$ by $-\epsilon H^{-1} \nabla L(z_i)/n$; chain rule $\to$ test-loss change $\mathcal{I}/n$.

Tricks: stochastic HVP (LiSSA), CG, never invert H. **Issues:** nonconvex $\to$ H not PD (add damping λI); deep-net influence noisy/unstable; fragile under retraining (Basu).

Other methods:

- **TracIn**: $\sum_{\text{ckpts}} \eta_t \nabla L(z_{\text{test}})^\top \nabla L(z_i)$. First-order, uses checkpoints.
- **Datamodels / TRAK**: train many models on random subsets, regress on subset indicator. Expensive but faithful.
- **Representer point**: last-layer kernel decomposition.

L18: Neurosymbolic Learning

Idea: neural perception + symbolic reasoning (logic, programs, KGs). Neural handles noise; symbolic handles structure, interpretability, constraints, compositionality.

Motivation: sample efficiency, OOD generalization, encode domain knowledge, interpretable intermediate symbols, guaranteed constraint satisfaction.

Examples:

- **DeepProbLog**: ProbLog + neural predicates. MNIST classifier is neural; $\text{sum}(X, Y, Z) \leftarrow Z = X + Y$ is symbolic.
- **Neural theorem provers**: learn proofs over KGs.
- **Scallop**: differentiable Datalog + provenance semirings.
- **Logic Tensor Networks**: fuzzy logic relaxations of FOL.
- **Neuro-symbolic concept learner**: objects + attrs + rels.

Key challenge: backprop through discrete ops (argmax, bool, count). **Solutions:** fuzzy/soft logic, REINFORCE (score-fn estimator), straight-through, Gumbel-softmax, semiring diff. (exact for Datalog).

Costs: hard to train (discrete bottlenecks), need domain knowledge to write rules, scalability, symbolic inference slow.

L19: RL Background

MDP: $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$. Policy $\pi(a \mid s)$. Trajectory $\tau = (s_0, a_0, r_0, \dots)$.

Value:

$$V^\pi(s) = \mathbb{E}_\pi\Big[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s\Big],\ Q^\pi(s,a) = \mathbb{E}[\cdot \mid s_0 = s, a_0 = a]$$

Bellman: $V^\pi(s) = \mathbb{E}[r + \gamma V^\pi(s')]$; $V^*(s) = \max_a \mathbb{E}[r + \gamma V^*(s')]$; $\pi^*(s) = \arg\max_a Q^*(s,a)$.

Objective: $J(\pi) = \mathbb{E}_{\tau\sim\pi}[\sum_t \gamma^t r_t]$.

Policy gradient (REINFORCE):

$$\nabla_\theta J = \mathbb{E}[\nabla_\theta \log \pi_\theta(a \mid s)\, A(s,a)]$$

$A = Q - V$ reduces variance vs raw return.

Algorithms:

- **REINFORCE**: MC PG, high variance.
- **Actor-Critic**: learn V (critic) + π (actor).
- **TRPO/PPO**: constrain π_{new} close to π_{old}; PPO clipped $\min(r_t A, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)A)$.
- **Q-learning / DQN**: off-policy value-based.
- **SAC**: off-policy AC + entropy bonus.

Online vs Offline:

- **Online**: agent interacts w/ env, fresh samples from π_θ.
- **Offline**: fixed dataset, no new samples. Risk: extrapolation, Q-overestimation on OOD actions. Fixes: BCQ, CQL, IQL.

On-policy vs off-policy: samples from current π (needed each update) vs any behavior policy (reuse buffer).

Occupancy:

$$d^\pi(s,a) = (1-\gamma)\sum_{t=0}^{\infty} \gamma^t \Pr_\pi(s_t = s, a_t = a)$$

Denoted $\mathcal{D}^\pi$ in RLHF objective.

Exploration: ϵ-greedy, UCB, entropy bonus, Thompson sampling.

L20: RLHF + DPO

3-stage pipeline:

1. **SFT**: imitate human demos, CE on (x, y_{expert}). Offline, supervised, no RL.
2. **Reward model** r_ϕ: preference pairs (x, y_w, y_l) via Bradley-Terry:
$$P(y_w \succ y_l) = \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))$$
Loss: $-\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))$.
3. **RL step** (PPO):
$$\max_\theta \mathbb{E}_{x\sim D, y\sim\pi_\theta(\cdot|x)}[r_\phi(x,y)] - \beta\, \text{KL}(\mathcal{D}^\pi \| \mathcal{D}^{\pi_{\text{ref}}})$$
Online samples from π_θ, online RL. Labels *offline* (baked into r_ϕ).

β trade-off (SA #5):

- $\beta \to 0$: no KL anchor $\to$ **reward hacking** / over-optimization. Policy exploits r_ϕ errors, produces degenerate outputs (repetition, gaming), loses coherence. Also π_θ drifts far from π_{ref}'s distribution $\Rightarrow r_\phi$ unreliable there.
- $\beta \to \infty$: policy pinned to $\pi_{\text{ref}} \to$ **no learning**, ignores reward, no improvement over SFT.

DPO: skip explicit reward+RL. Closed-form solution:

$$\pi^*(y \mid x) = \tfrac{1}{Z(x)}\pi_{\text{ref}}(y \mid x)\exp\big(\tfrac{1}{\beta} r(x,y)\big)$$

$\Rightarrow r(x,y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$. Plug into BT ($Z(x)$ cancels):

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}\Big[\log \sigma\Big(\beta \log \tfrac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \tfrac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\Big)\Big]$$

Just SGD on preference data. No reward model, no sampling during training, more stable than PPO-RLHF.

RLHF table (MCQ #5):

Alg	Onl.samp	Onl.lab	Onl.RL	Off.RL
SFT	F	F	F	F (not RL)
Model-RL	T	F	T	F
DPO	F	F	F	T

Key: "online labels" = new human feedback *during* training (none; preferences pre-collected). SFT is supervised, not RL. DPO trains on fixed preference dataset $\to$ offline, no sampling from π_θ.

Recourse Example (SA #1)

Scenario: Bank denies Alice a loan. Explainer (LIME/SHAP/CF) shows: "debt-to-income above 0.45 drove denial; 0.35 flips it." Alice has **actionable recourse**—pay debt or raise income—to get approved. Explanation turns opaque "no" into action plan. *Others:* college admission ("GPA +0.2 flips"), parole ("complete program X"), medical ("A1C ¿7 $\to$ diet/meds"), hiring ("add cert Y"), insurance ("reduce mileage").

Quick-Recall Tips

- **Efficiency**: $\sum \phi_i = f(x) - \mathbb{E}[f]$. Don't confuse with Symmetry.
- **Input gradient**: sensitivity of *output logit* w.r.t. pixels.
- **LOO impractical**: *computational cost*, not impossibility.
- **Manhattan CF**: *same scale*, not independence.
- **KL**$\to 0$: reward hacking; $\to \infty$: no learning.
- **SHAP marginal**: fast but breaks under correlation.
- **LIME local fidelity**: $g \approx f$ *near* x, not globally.
- **IG completeness**: $\sum$ attribs $= f(x) - f(\text{baseline})$.
- **Shapley 4 axioms**: Efficiency, Symmetry, Null, Linearity.
- **TCAV**: concept dir from linear probe on activations.
- **DPO**: closed-form of RLHF; $Z(x)$ cancels pairwise.

Method Comparison

Method	Scope	Needs	Output
LIME	local	query	feat weights
SHAP	local	query	attributions
Saliency	local	grad	pixel heatmap
IG	local	grad+base	pixel attrib
Grad-CAM	local	conv acts	coarse heatmap
TCAV	global	activations	concept score
CF	local	query	alt input x'
Influence	global/local	Hessian	train pt score
TracIn	global/local	grads+ckpts	train pt score

Long Answer: LASSO-LIME Duplicates

Setup: features i, j satisfy $X_i = X_j$. Objective:

$$\hat\beta = \arg\min_\beta \sum_k w_k(y_k - x_k^\top \beta)^2 + \lambda\|\beta\|_1$$

(a) Prediction depends on β_i, β_j only through $s = \beta_i + \beta_j$.

Contribution of features i, j to prediction on row k:

$$X_{ki}\beta_i + X_{kj}\beta_j = X_{ki}(\beta_i + \beta_j) = X_{ki} \cdot s$$

since $X_{ki} = X_{kj}$. For *every* row k, the fitted value $x_k^\top \beta$ is unchanged when (β_i, β_j) is replaced by any pair with same sum s. Thus the squared-error $\sum_k w_k(y_k - x_k^\top \beta)^2$ depends on (β_i, β_j) only through s. $\square$

(b) Minimize $|\beta_i| + |\beta_j|$ s.t. $\beta_i + \beta_j = s$.

Claim: minimum is $|s|$, achieved iff β_i, β_j have the **same sign** (zero compatible with either).

Proof (triangle inequality):

$$|\beta_i| + |\beta_j| \geq |\beta_i + \beta_j| = |s|.$$

Equality condition: tight iff both ≥ 0 (when $s \geq 0$) or both ≤ 0 (when $s \leq 0$). Any split preserving same sign works: $(s, 0)$, $(0, s)$, $(s/2, s/2)$, etc.

Consequence: combining (a)+(b), the full objective at any same-sign (β_i, β_j) with $\beta_i + \beta_j = s^*$ is identical. Solution is **non-unique**—a whole segment of optimizers exists. This is the classic "LASSO arbitrarily picks one of correlated/duplicated features" phenomenon; attributions become unstable (bad for explanations).

Fix: Elastic net adds ℓ_2: $\lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2$. $\beta_i^2 + \beta_j^2$ is *strictly convex*, minimized at $\beta_i = \beta_j = s/2$ on $\{\beta_i + \beta_j = s\}$, breaking the tie and spreading mass *evenly*.

Common Gotchas

- **Null Player**: $\phi_i = 0$ requires $v(S \cup \{i\}) = v(S)$ for *every* S.
- **DPO**: $Z(x)$ cancels in pair-differences; never compute it.
- **PPO-RLHF KL**: on distribution $\mathcal{D}^\pi$ (implementations often approximate via per-token KL reward).
- **LIME**: fit on interpretable $z' \in \{0,1\}^{d'}$ but query f on raw z.
- **IG baseline** x' matters: black vs blur vs mean $\to$ different attributions.
- **Influence** assumes $\hat\theta$ minimizes convex loss; deep nets violate this.
- **CF** flips prediction; doesn't explain *why* f classified x that way.
- **LIME instability**: random perturbations $\Rightarrow$ reruns differ. Fix: more samples, fixed seed.
- **Shapley complexity**: exact is $O(2^n)$; use KernelSHAP / TreeSHAP.
- **SHAP baseline** is $\mathbb{E}[f(X)]$ (scalar), unlike IG baseline (input).
- **Grad-CAM resolution**: limited by last conv layer (7×7 ResNet). Upsample.